TRAINING AID

The DRS sheets, broken down rule by rule

Every Fluidstack **Deployment Requirement Sheet** the course covers so far, restated in plain words with drawings: what each rule means, why it exists, and where it is taught. Sheets covered: 01.00.00, 01.00.01, 01.00.02, 01.00.03 and 01.00.03.01 (revisions of 6/9/2026).

> **The current DRS is the requirement.** This page is a study aid built from the DRS sheets added to the course so far. Check the revision date against the Fluidstack portal, have the actual DRS on hand where you work, and where anything is unclear or disagrees: **ask your lead.**

The five sheets on this page

`01.00.00` Introduction: what a DRS is

`01.00.01` General Fiber Cable Dressing

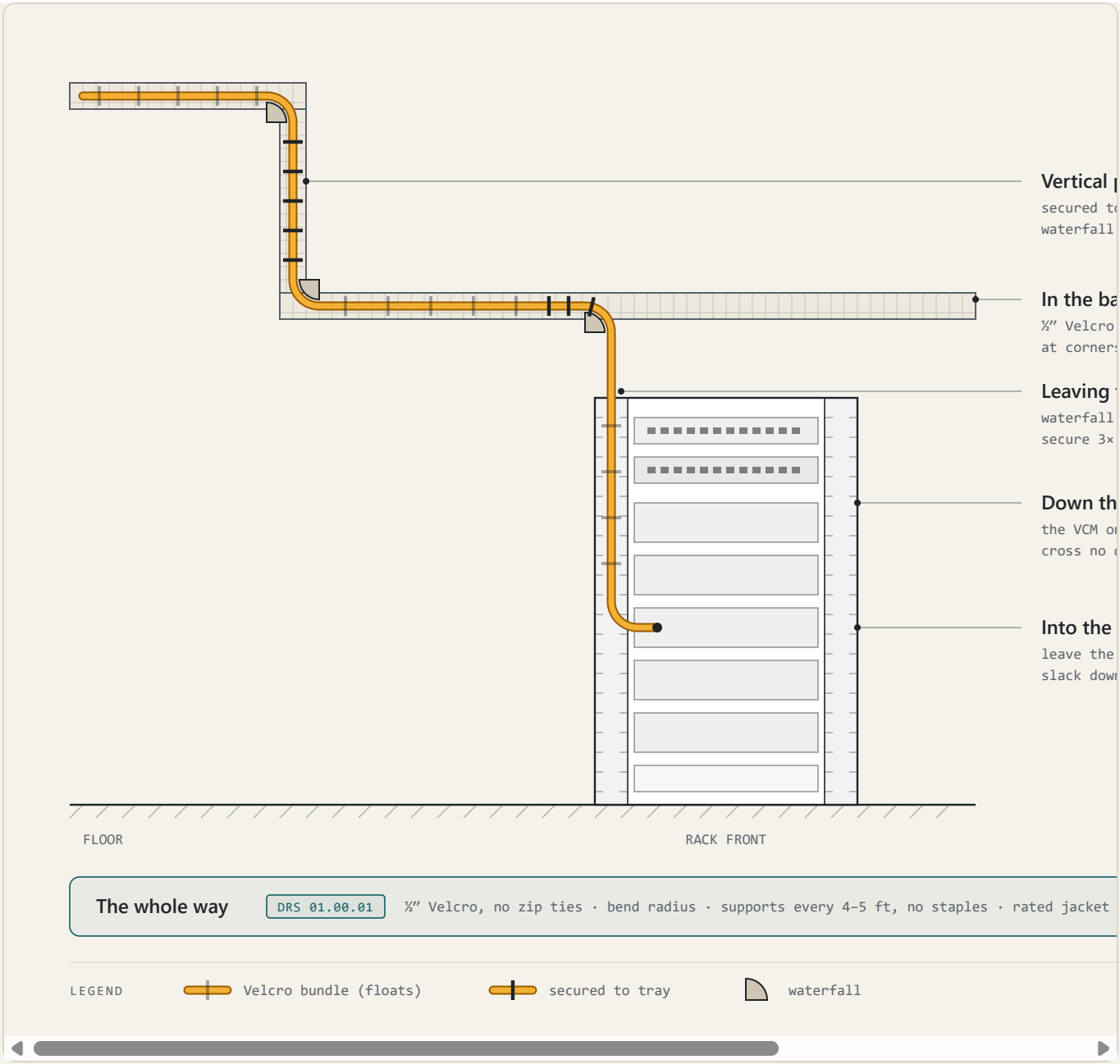
`01.00.02` Front of Rack Cable Dressing

`01.00.03` Cable Dressing Basket Trays

`01.00.03.01` Cable Dressing Vertical Pathways

One cable, four sheets

Before the rule-by-rule pages, see how the dressing sheets divide one cable's route. The general rules (01.00.01) apply the whole way; the tray rules (01.00.03) apply in the basket; vertical pathways (01.00.03.01) take over where the run turns down; and the front of rack (01.00.02) governs the last foot into the device.



One route from an upper tray to the device, with the sheet for each part. Grey ticks: Velcro bundles that float in the tray. Black ticks: ties to the tray. Every sheet is broken down rule by rule in DRS Explained.

01.00.00

Introduction: what a DRS is

Portal page · approved 6/9/2026 · approved by Tom Marsh, Infrastructure Deployment Manager · **Internal Use Only**

This is not a task sheet. It is the portal page that explains what a DRS is, where it lives, and the rule that you must have it on hand. Read it once, carefully — every other sheet assumes it.

The rules, one by one

R1 DRS = Deployment Requirement Sheet: Fluidstack's technical standards for how installation work must be done. **Requirements, not guidelines.** Every on-site task has a matching DRS.

In plain words. A guideline is advice; a requirement is the way it must be done, every time. If a coworker, a photo from another site, or this course says something different, the DRS wins.

R2 Contains step-by-step requirements, specifications, approved materials, diagrams, and references to ANSI/TIA, BICSI, NEC and NEMA standards.

In plain words. You do not need to memorize the standards themselves. The DRS turns them into the rules you follow — but recognize the names, because they appear on drawings, cable jackets and in conversations.

R3 Exists so every deployment meets the same quality standard, whoever does the work; protects equipment and lowers outage risk.

In plain words. The same job done two different ways is how outages happen. The DRS is what lets any qualified person pick up the work and know exactly how it must be done.

R4 Approved by Fluidstack ICT leadership. New revisions are announced through the portal. **Always use the current version: check the revision date on the cover page.**

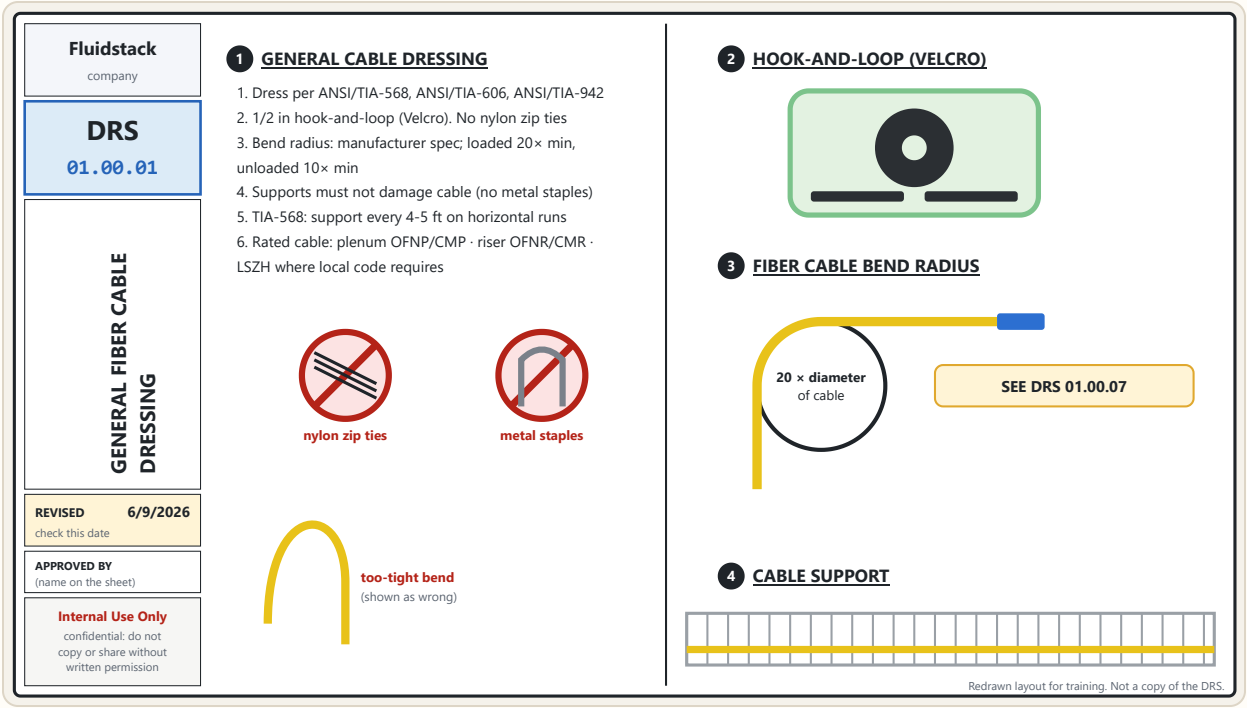
In plain words. Standards change; when one changes, a new DRS revision is issued. An old printout is the classic mistake: before every task, check the revision date against the portal.

R5 Get it from Fluidstack Confluence (fluidstack.atlassian.net → LVC Deployment space → DEPLOYMENT REQUIREMENTS SHEETS). Backup: your Fluidstack PM or site supervisor.

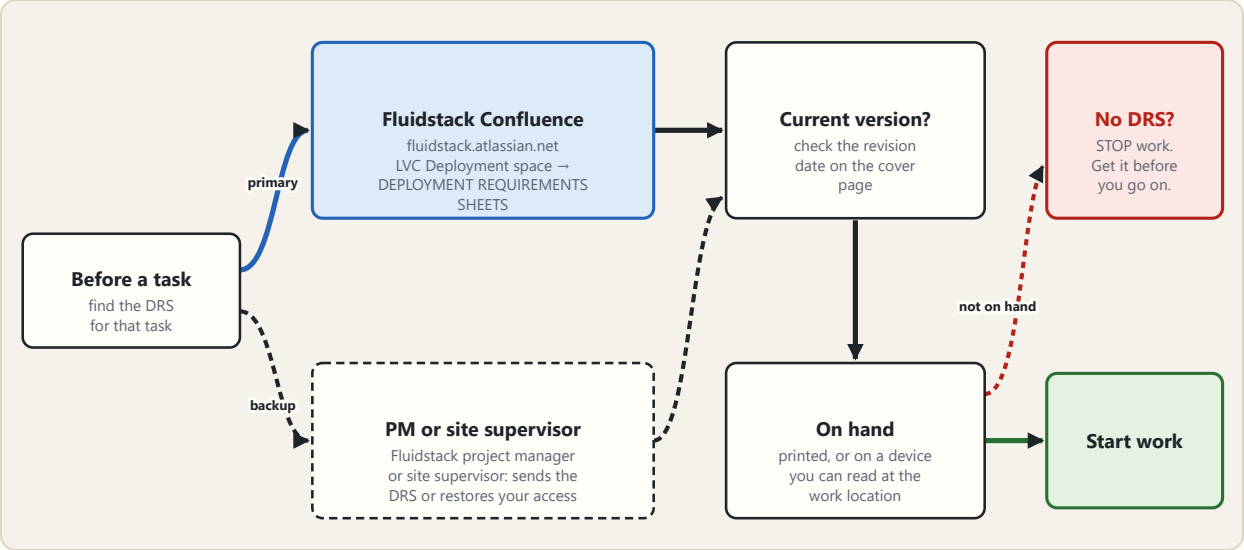
In plain words. The portal needs a login and a connection, and data hall floors have dead zones. Get or print the DRS for the day's tasks before you go out on the floor.

R6 **On site:** have the applicable DRS on hand, printed or on a device. Working from memory alone is not acceptable. No DRS: **stop work and get it.**

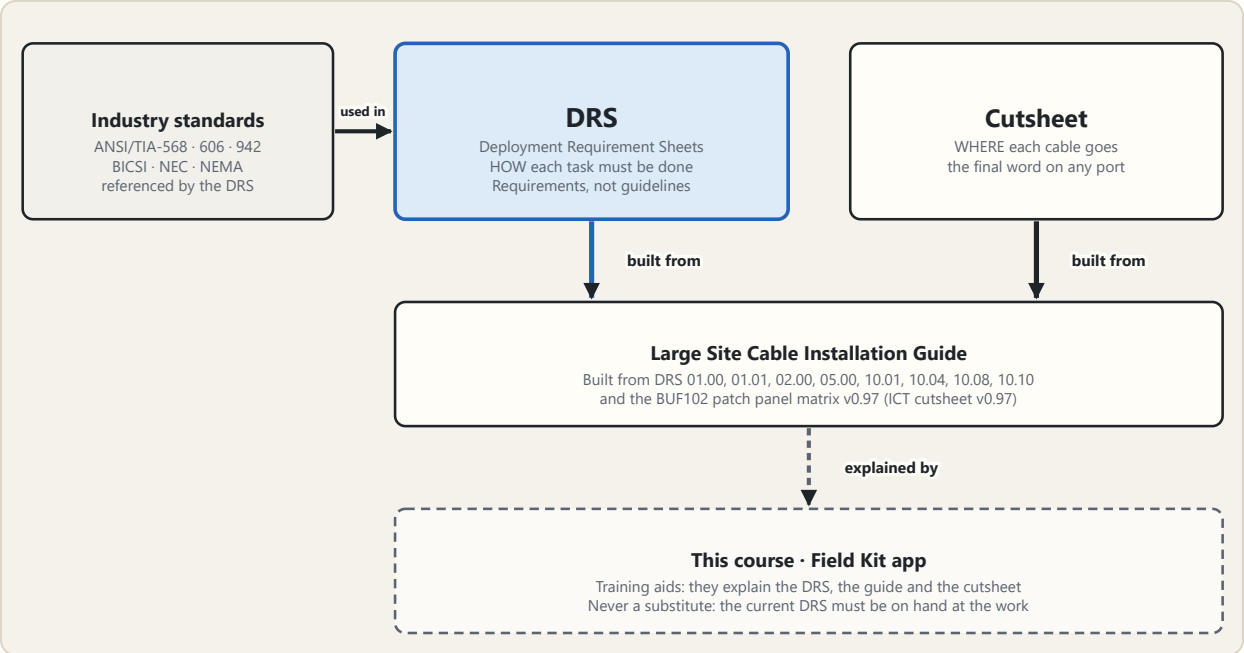
In plain words. This is the on-site rule. 'On hand' means at the work location — printed, or on a device you can actually read. If you cannot, you stop and get it before you continue.



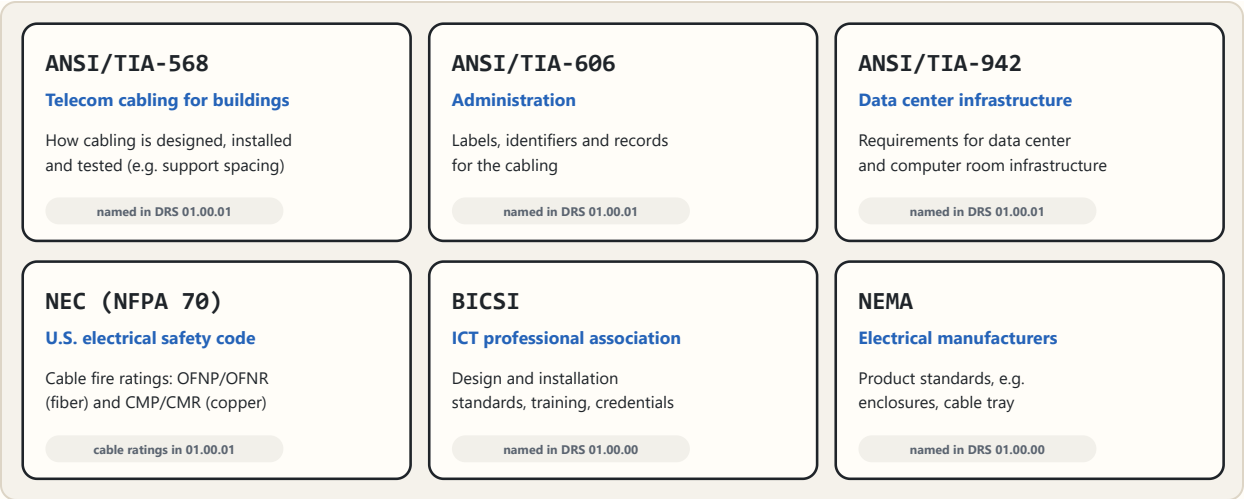
Anatomy of a DRS sheet: every sheet has a number, a title, a revision date, an approval block, the Internal Use Only marking, numbered rules, crossed-out 'never' pictures, and cross-references to other sheets.



How you get the DRS: the Confluence portal first; your Fluidstack PM or site supervisor as backup. Then check the version before you rely on it.



How the DRS fits with the other documents: the DRS says how, the cutsheet says where (final word on any port), the installation guide was built from the DRS, and this course and the Field Kit are training aids that never replace it.



The industry standards named in the DRS sheets and what each covers — TIA-568 (telecom cabling), TIA-606 (administration: identifiers and labels), TIA-942 (data center infrastructure), NEC (fire ratings), BICSI (methods), NEMA (equipment).

Standards named
ANSI/TIA, BICSI, NEC, NEMA

Points to
None.

Taught in
Lesson 0 · What a DRS is · Lesson 0 · Getting the right DRS · Lesson 0 · The on-site rule

Video
Lesson 0 video

Open items 14, 17 — see the Reference guide, *Ask your lead: open items*.

01.00.01

General Fiber Cable Dressing

portal title: General Cable Dressing

Drawing + portal summary · revised 6/9/2026 · approved by Tom Marsh · **Internal Use Only**

The six general rules for dressing any cable on this project. Every task sheet builds on these, and the bend-radius numbers here are the ones people get wrong under time pressure.

The rules, one by one

R1 Dress cables in accordance with **ANSI/TIA-568, ANSI/TIA-606 and ANSI/TIA-942**.

In plain words. The DRS names the standards so the requirement is traceable; on site you cite the DRS rule itself (for example: DRS 01.00.01, rule 2).

R2 Bundle with **1/2-inch hook-and-loop (Velcro)**. Nylon zip ties are not allowed.

In plain words. Zip ties crush fiber. 1/2-inch Velcro holds the bundle without pinching it. The crossed-out picture on the sheet is a zip tie — 'never'.

R3 **Bend radius:** follow the manufacturer's spec; minimum **20 × diameter loaded** (under tension), **10 × unloaded**. See DRS 01.00.07 for fiber bend radius.

In plain words. Loaded means while you are pulling or the cable carries its own weight; unloaded means it is resting dressed. Worked example: a 1-inch trunk bundle needs a 20-inch radius while you pull it. Tighter than that and you crack glass you cannot see broken.

R4 Cable supports must not damage the cable: **no metal staples**.

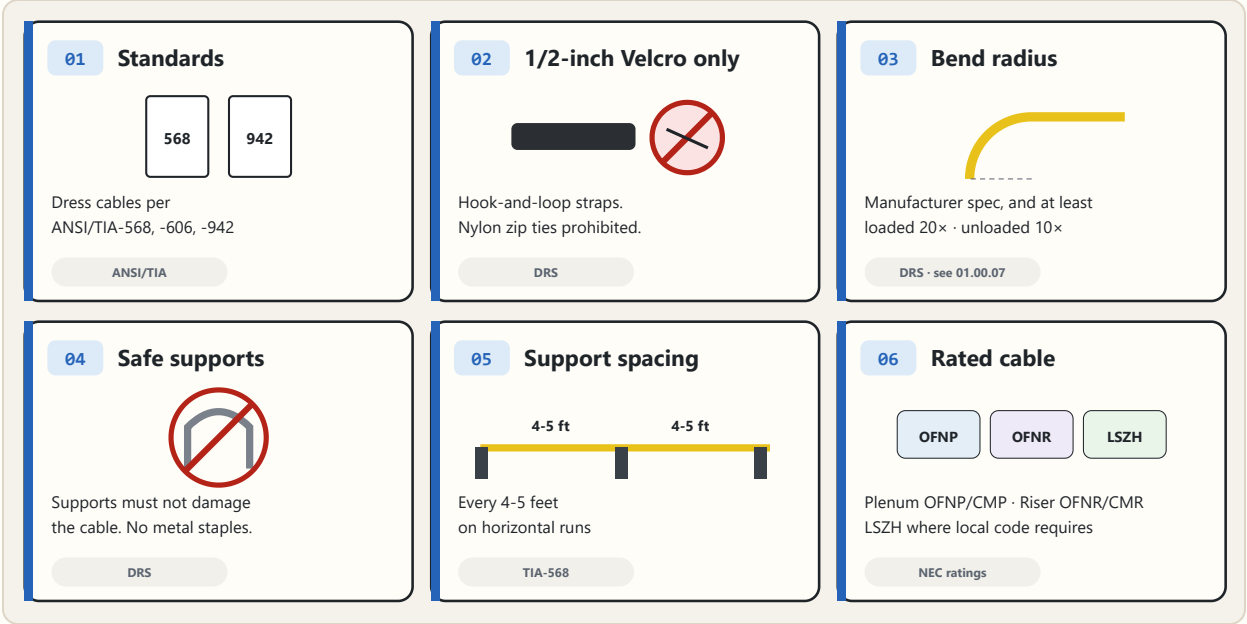
In plain words. Anything that pinches or pierces the jacket is damage. Supports hold the cable; they never bite into it.

R5 **Support every 4-5 feet** on horizontal runs (TIA-568 recommendation).

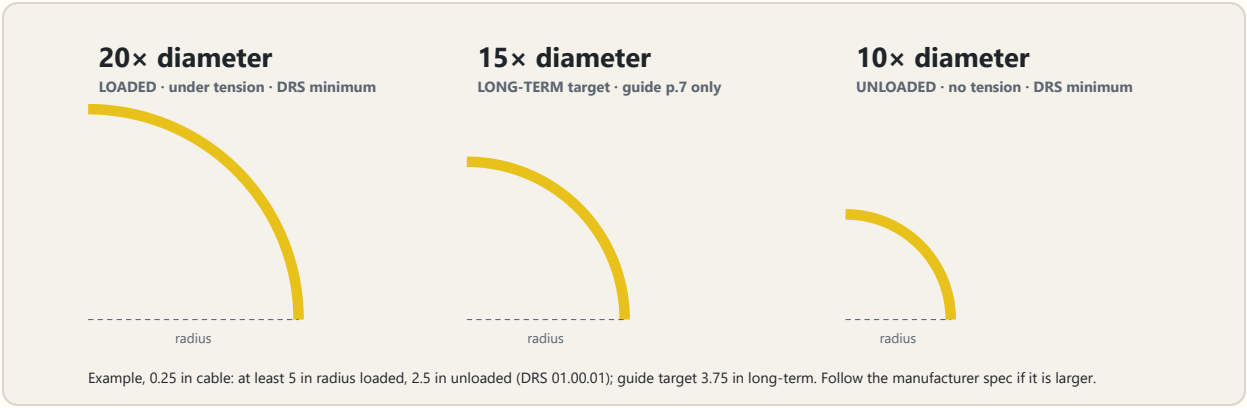
In plain words. Unsupported spans sag; sagging cable stresses the jacket and the fibers at every high point. Pace it out: a support about every long stride.

R6 **Rated cable for the space:** plenum OFNP / CMP in plenum spaces, riser OFNR / CMR in vertical risers, LSZH where local code requires.

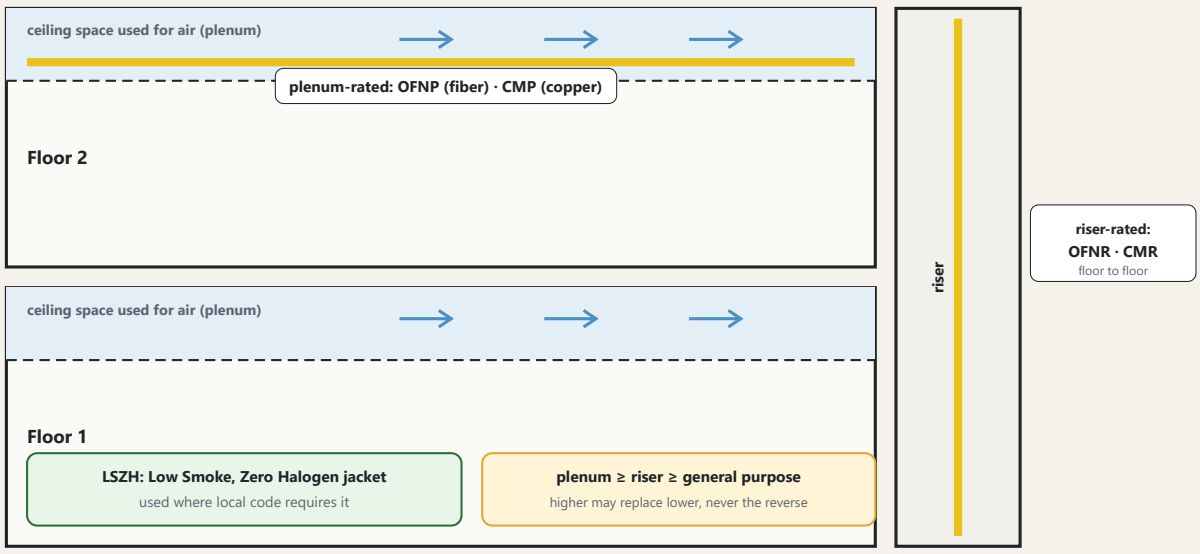
In plain words. The rating is printed on the jacket. The wrong rating in a plenum or riser is a code violation the fire marshal can fail — it is not a dressing detail.



The six rules on one drawing: Velcro not zip ties, bend radius, no damaging supports, support spacing, rated cable for the space.



Bend radius, worked: 20 × diameter loaded (under tension), 10 × unloaded — and what a too-tight bend does.



Jacket ratings and where each belongs: OFNP/CMP in plenums, OFNR/CMR in risers, LSZH where local code requires.

Standards named
ANSI/TIA-568, ANSI/TIA-606, ANSI/TIA-942, NEC (cable ratings)

Points to
DRS 01.00.07

Taught in
Lesson 5 · The six rules · Lesson 5 · The standards behind the rules · Lesson 5 · Ties, bends and supports · Lesson 5 · Rated cable · Lesson 5 · Bend radius worked example · Lesson 3 · Handling fiber

Video
Lesson 5 video; Lesson 3 video (Handling fiber)

Open items 12, 13, 15, 16 — see the Reference guide, *Ask your lead: open items*.

01.00.02

Front of Rack Cable Dressing

Drawing + portal summary · revised 6/9/2026 · approved by Tom Marsh · **Internal Use Only**

How cables must arrive at the rack: where they leave the tray, how they drop, and the replaceable-device rule that drives everything else in the rack.

The rules, one by one

R1 Cables shall **not cross devices they do not patch into**. Every device must be replaceable without unpatching cables that end in another device.

In plain words. The replaceability rule. If a cable crosses over a device it has nothing to do with, swapping that device means unpatching someone else's cable — that is an outage you caused. Cables land on their side, in their waterfall, into their VCM.

R2 Cable shall **exit the basket tray directly above the rack edge or the vertical cable managers** (VCMs).

In plain words. The drop starts in the right place: straight above the rack edge or the VCM, not half a bay away where the cable would angle across neighbors.

R3 All cables entering or leaving the basket tray must be **supported by waterfalls** to keep the proper bend radius.

In plain words. A waterfall is the smooth curved exit accessory at the tray edge. Cable hanging over a bare tray edge bends tighter than radius and scores the jacket.

R4 **Over-the-top waterfalls** must be used for **network racks**.

In plain words. Network racks are the exception with their own pattern: cords come over the top. If you see networkrack traffic exiting sideways, stop and check the sheet.

R5 Bundle cable neatly with **1/2-inch Velcro**.

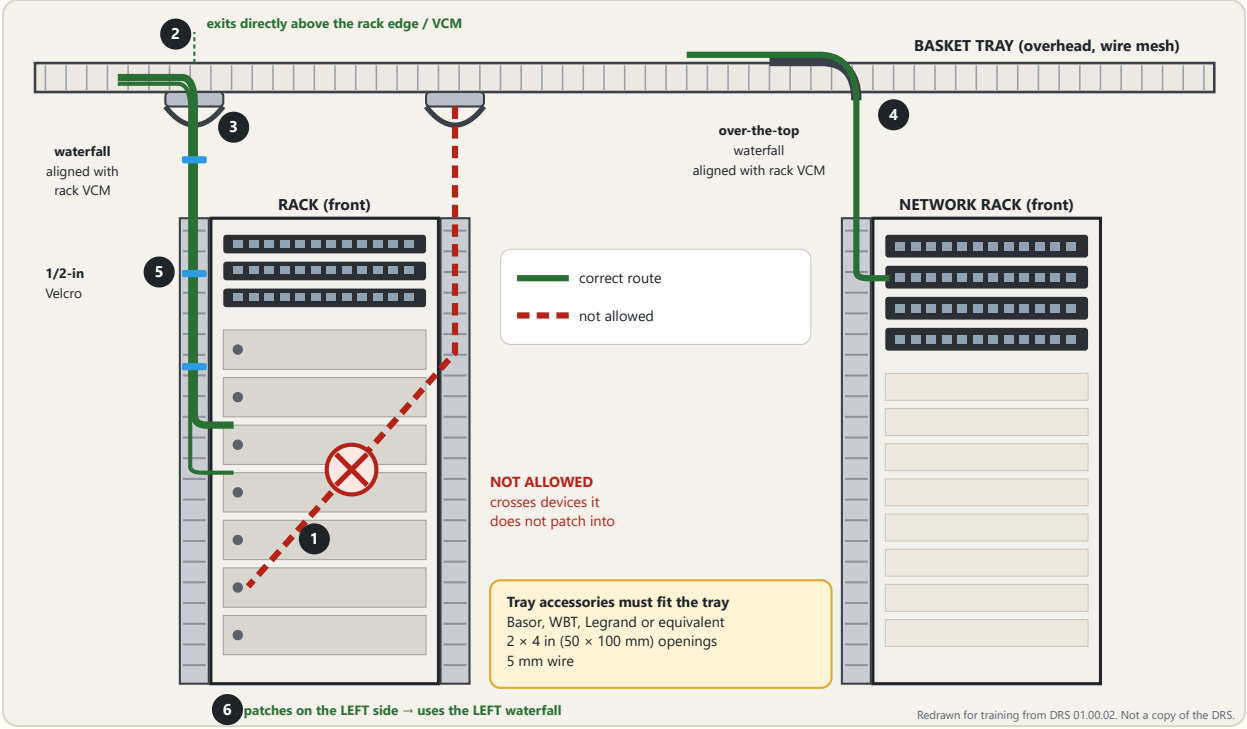
In plain words. Same tie rule as 01.00.01, restated where the work happens: 1/2-inch hook-and-loop, never zip ties.

R6 Cables should use the **waterfall next to the side of the rack they patch into**.

In plain words. Left-side ports use the left waterfall, right-side ports the right one. Crossing under or over to the far side tangles the rack face.

R7 Drawing note: each waterfall lines up with the rack VCM. **Tray accessories must fit** trays made by Basor, WBT, Legrand or equivalent, with 2 × 4 inch (50 × 100 mm) openings and 5 mm wire.

In plain words. The hardware note from the drawing: waterfalls and tray accessories are sized for that basket (2×4-inch mesh, 5 mm wire). If the parts do not fit the tray as drawn, that is a question for your lead, not something to force.



Front of the rack, drawn: tray exits above the VCMs, waterfalls carrying each drop, bundles arriving at the rack face on their own side.

Standards named

None named on this sheet.

Points to

None.

Taught in
Lesson 5B · Front of the rack · Lesson 5B · Worked example · Lesson 9 · Landing the 192F (network rack) · Lesson 9 · Cords to the spines (network rack)

Video
Lesson 5B video (Front of the rack)

Open items 12, 18, 19 — see the Reference guide, *Ask your lead: open items*.

01.00.03

Cable Dressing Basket Trays

portal title: Basket Tray Cable Dressing

Drawing + portal summary · revised 6/9/2026 · approved by Tom Marsh · **Internal Use Only**

What 'neat' means inside the tray itself: how often to tie, where to anchor, and the corner-and-exit rule that keeps bundles in place without strapping them down everywhere.

The rules, one by one

R1 Bundle cables neatly in the tray with **Velcro 12 to 18 inches apart**.

In plain words. In the tray the Velcro rhythm is roughly every 12 to 18 inches — a hand span to a forearm apart. Tighter is not better; you are keeping the bundle tidy, not compressing it.

R2 **Do not secure bundles to the basket tray** except going around bends, exiting the tray, or running vertically. **Secure 3× at corners and 3× at exits.**

In plain words. Horizontal bundles float in the tray — they are not zip-tied to the mesh. They get anchored at the three places it matters: corners (three ties), exits (three ties), and vertical runs.

R3 **1/2-inch Velcro** for all cable bundling.

In plain words. The 1/2-inch rule again — the one constant across every dressing sheet.

R4 **Waterfalls centered over the rack VCMs.**

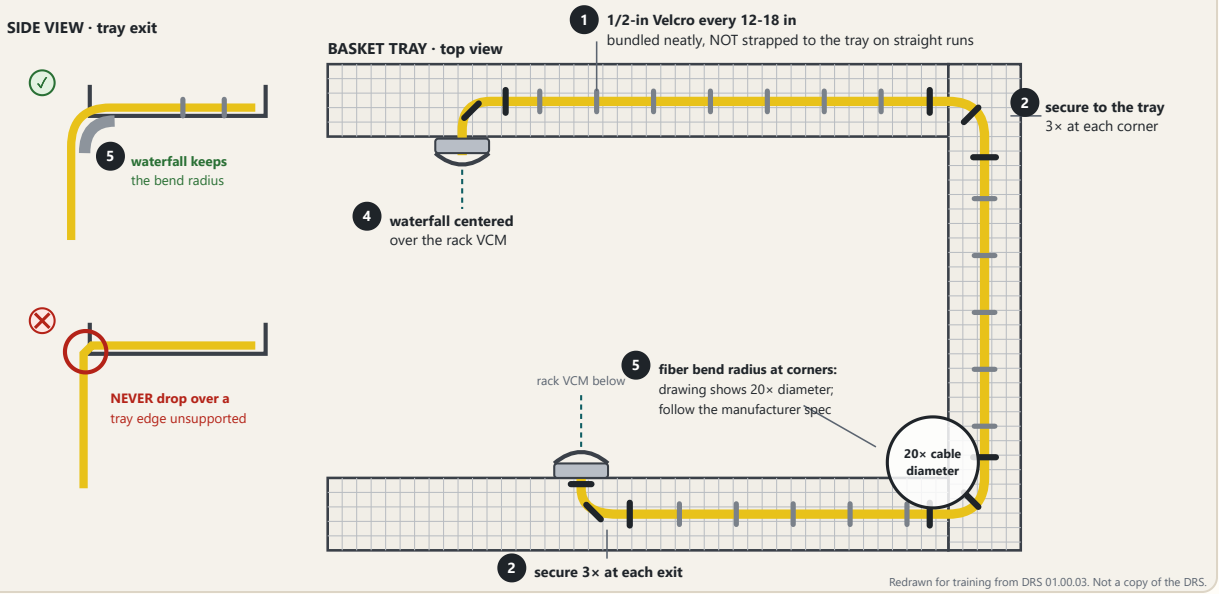
In plain words. Alignment rule: the waterfall sits centered over the vertical cable manager of the rack it feeds, so the drop goes straight down, not diagonal.

R5 Cables keep the **manufacturer's recommended bend radius** at all times. Drawing: fiber bend radius at a tray corner shown as **20 × cable diameter**.

In plain words. At a tray corner the bundle sweeps; the drawing sizes that sweep at 20 × diameter for fiber — the loaded number from 01.00.01, because a corner is where the cable carries load.

R6 Portal summary: **never allow cable to drop over a tray edge unsupported**. Waterfalls keep the bend radius at every tray exit.

In plain words. If there is no waterfall at an exit, the cable must not go over that edge. Get the waterfall installed — do not 'just this once' a bare-edge drop.



In the tray: Velcro every 12–18 inches, bundles floating on the horizontal run, anchored 3× at the corner and 3× at the exit, waterfall centered over the VCM below.

Standards named
None named on this sheet.

Points to
None.

Taught in
Lesson 5B · From the tray to the device · Lesson 5B · In the basket tray · Lesson 5B · Worked example · Lesson 11 · Pulling and dressing the OCS bundles · Lesson 12 · Routing and labels

Video
Lesson 5B video (In the basket tray); Lesson 11 video (Pulling and dressing)

Open items 12, 20 — see the Reference guide, *Ask your lead: open items*.

01.00.03.01

Cable Dressing Vertical Pathways

portal title: Vertical Pathway Dressing

Drawing + portal summary · revised 6/9/2026 · approved by Tom Marsh · **Internal Use Only**

The companion sheet to 01.00.03 for vertical tray: gravity now works on the cable, so the rules change from 'float' to 'hold' — self-supporting bundles, waterfalls at every transition.

The rules, one by one

R1 Bundle cables neatly with **Velcro 12 to 18 inches apart**, in vertical tray as well.

In plain words. Same rhythm as the horizontal tray, now running up the wall or between trays.

R2 **Secure bundles to the cable tray when running vertically** (horizontal bundles float).

In plain words. This is the flip of the 01.00.03 rule: vertical bundles ARE secured to the tray, at the 12–18 inch rhythm, so the cable cannot slide or slump.

R3 Cable in vertical tray or ladder must be **secured to be self-supporting**. Excessive cable weight on horizontal transitions damages the tray and the cable.

In plain words. Self-supporting means the vertical run carries its own weight at its own anchors — the weight must never hang on a horizontal tray transition at the top or bottom. That is how trays and cables get damaged.

R4 **Waterfalls are required at all vertical transitions** to keep the cable bend radius. A missing waterfall can damage the cable.

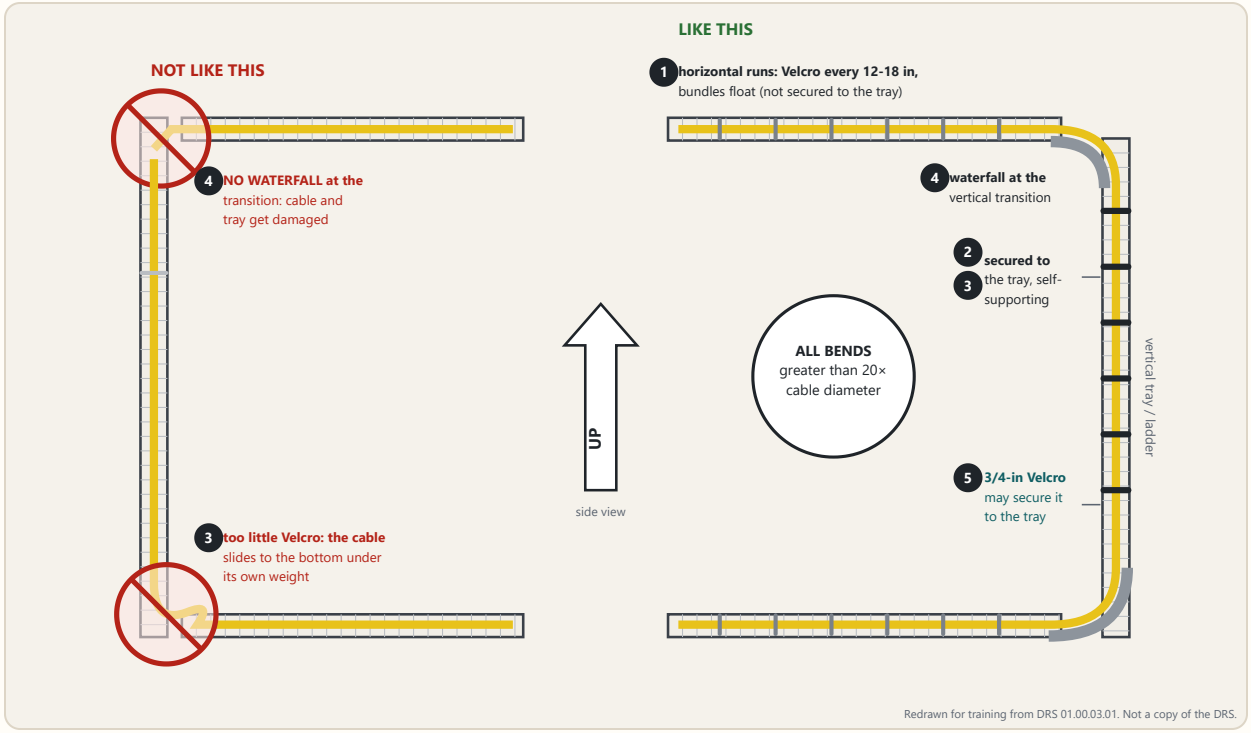
In plain words. Every place the run changes between horizontal and vertical, the bend goes through a waterfall. 'All' means all — this is the most-cited rule on vertical work.

R5 **3/4-inch Velcro may be used to secure cable to vertical tray.**

In plain words. The one size exception: 1/2-inch for bundling everywhere, 3/4-inch allowed where you are anchoring the bundle to the vertical tray.

R6 Drawing: **all bends greater than 20 × the cable diameter**. Portal: **too little Velcro on vertical runs lets cable drop to the bottom** under its own weight, damaging cable and tray.

In plain words. Two failure pictures in one rule: a bend tighter than 20× at the transition, or a sparse-tied run where the cable slides down and piles its weight on the bottom.



Vertical pathways: bundles secured to the vertical tray and self-supporting, waterfalls at every transition, all bends kept beyond 20 × diameter.

Standards named
None named on this sheet.

Points to
None.

Taught in
Lesson 5B · From the tray to the device · Lesson 5B · Vertical pathways · Lesson 5B · Worked example · Lesson 5 · Velcro sizes · Lesson 11 · Pulling and dressing the OCS bundles

Video
Lesson 5B video (Vertical pathways)

Open items 12, 20 — see the Reference guide, *Ask your lead: open items*.

next

Referenced, not yet in the course

Tracked as open items in the Reference guide

`01.00.07` Fiber bend radius. DRS 01.00.01 says "see DRS 01.00.07" — the dedicated bend-radius sheet is not in the course yet. Until it is added, use the 01.00.01 numbers (20× loaded, 10× unloaded) and ask your lead for the sheet when the work is specifically about bend radius.

`01.00, 01.01, 02.00, 05.00, 10.01, 10.04, 10.08, 10.10` The DRS sections the installation guide says it was built from (guide p.1). Which sheets belong to each section is not stated yet — tracked as an open item in the Reference guide.

When a new DRS sheet is added to the course, it gets its own page here and an entry on the course's DRS sheets list.

DRS Explained · Data Hall 4 Cable Installation training package · built from the Fluidstack DRS sheets covered by the course (revisions of 6/9/2026) and the Large Site Cable Installation Guide (cover Rev D, footers Rev C, 10/8/2026). **Internal Use Only:** keep within the project; do not post publicly.

Training aid only — never a substitute for the current DRS from the portal or your site's cutsheet.